

SPEECH INFORMATION RETRIEVAL

Speech recognition has the potential to dramatically change the way we create, capture, and store knowledge

MODEL

Voice -> Text

Keyword

BoW
Occurrences of words within a document

Bag of Words
(BoW)

Pattern Matching

Voice based reply

no

if matches
with database

yes

Voice Reply

ACCURACY METRICS

1. Pruning
2. N-best hypothesis and Word-graph
3. Multiple pass decoding
4. Relevance ranking

STEP 1: VOICE TO TEXT

Fuzzy logics

STEP 2: PATTERN MATCHING

STEP 3: TEXT TO VOICE

REAL-LIFE APPLICATIONS

- Virtual assistants and chatbots
- Voice-activated search functions
- Text-to-speech engines
- In-car command prompts
- Transcriptions of meetings or calls
- Enhanced security
- Phone directories
- Translation services



CHALLENGES

- Highly spontaneous, unprepared speech
- BoW - vocabulary, sparsity, meaning
- Unknown content lacking support in general language model
- Wide variety of accents and speaking styles
- OOV (Out-of-Vocabulary) queries

FUTURE

- Multilingual models
- Improved accuracy
- Better Human-Machine collaboration

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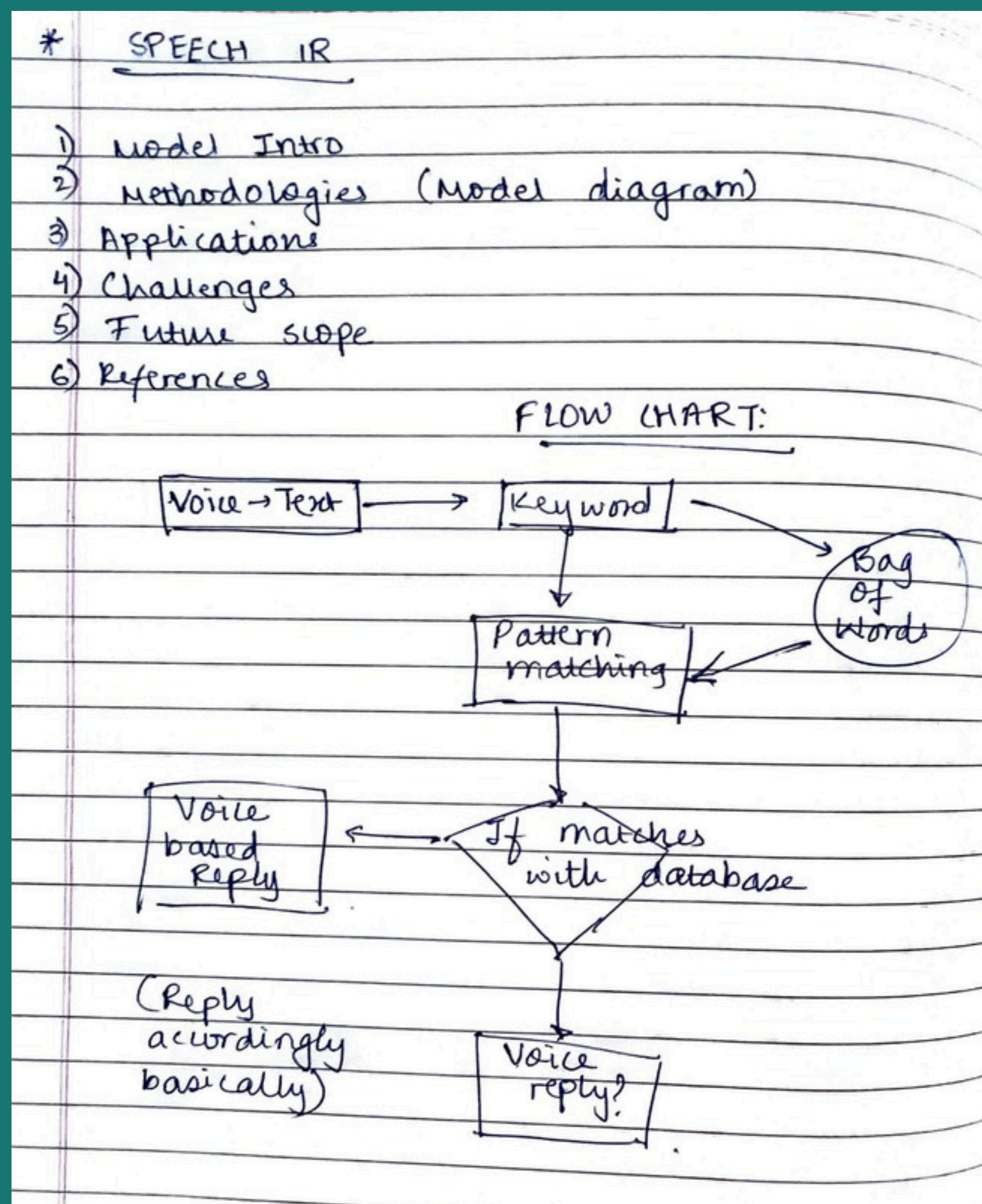
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Process

My brainstorming usually involves the following 4 steps:

1. Researching & noting content down

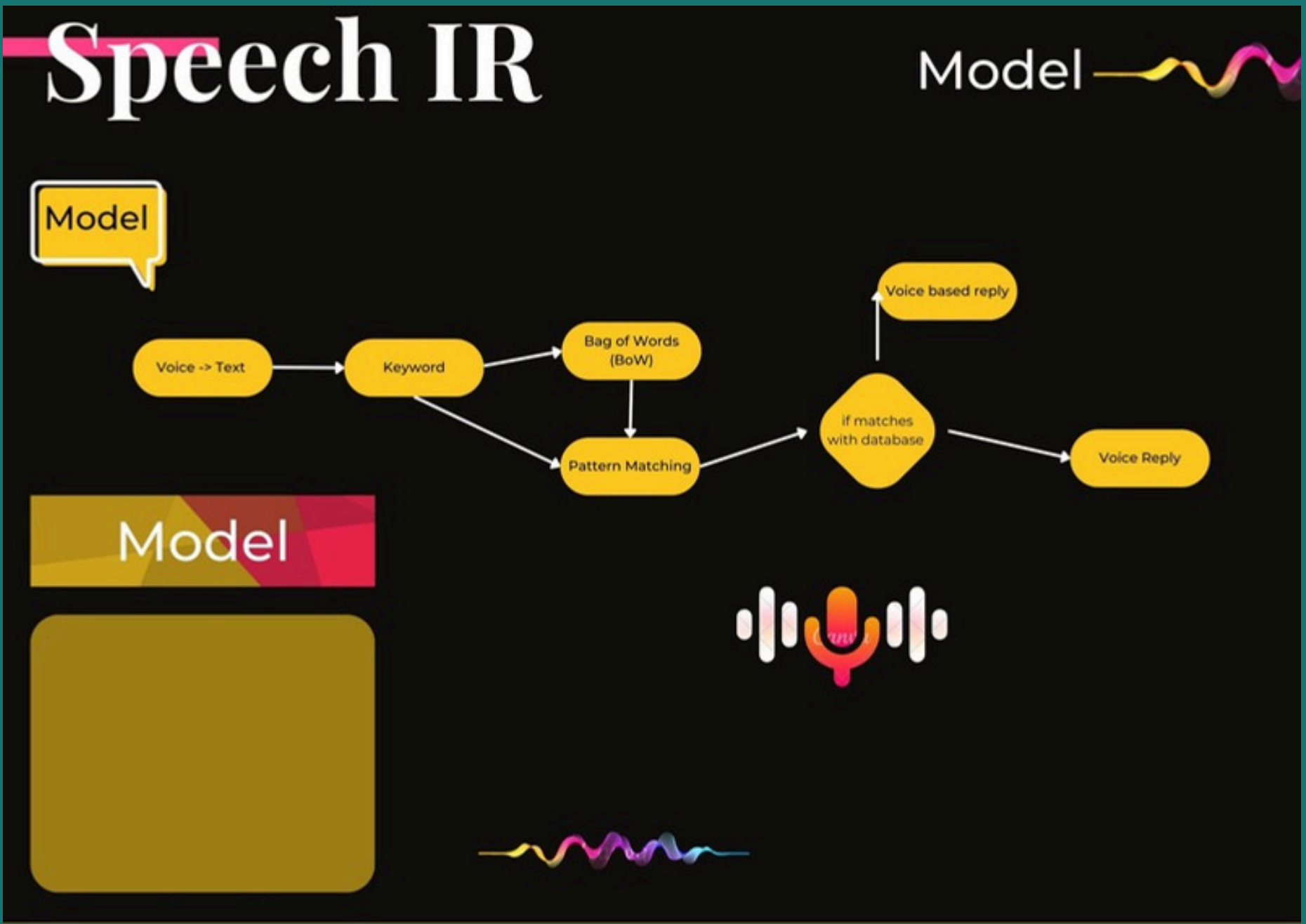
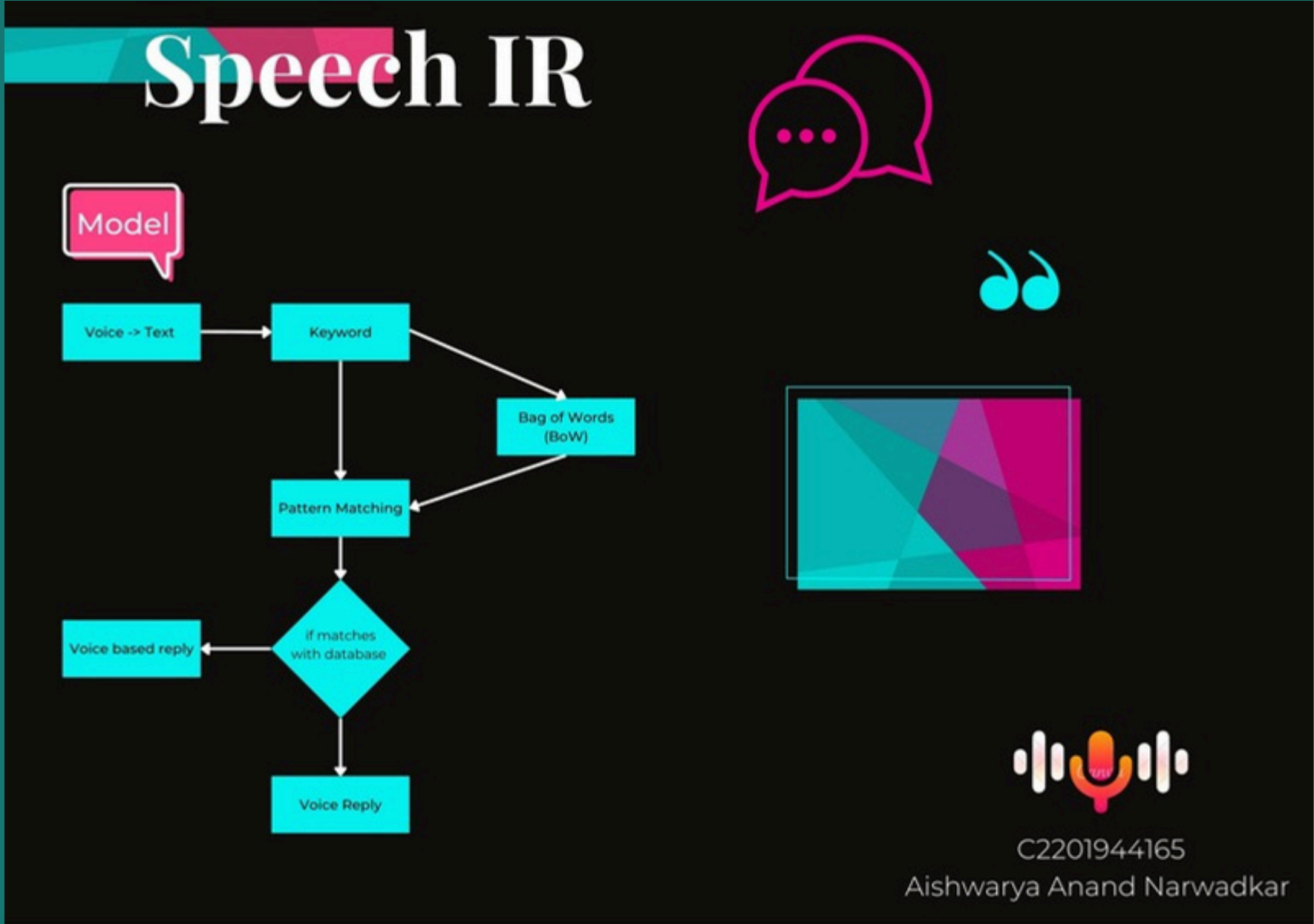


Virtual assistants and chatbots
Voice-activated search functions
Text-to-speech engines
In-car command prompts
Transcriptions of meetings or calls
Enhanced security with voice recognition
Phone directories
Translation services

Language is the foundation of these communications and, with the combination of AI, language processing, audio and speech technologies, companies can create more efficient, personalized customer experiences.

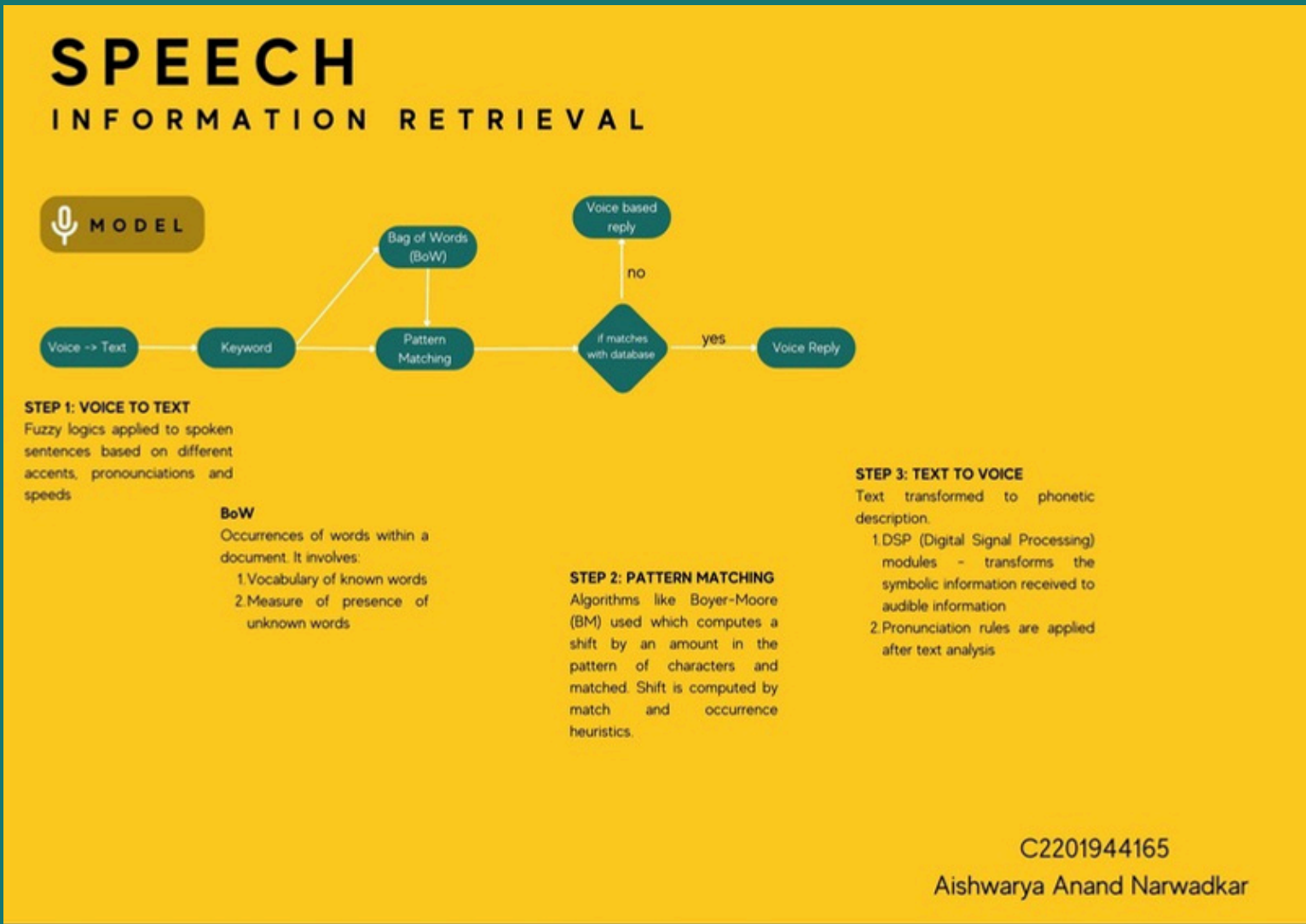
In machine learning, audio analysis includes a wide range of technologies: automatic speech recognition, music information retrieval, auditory scene analysis for anomaly detection, and more. Models are often used to differentiate between sounds and speakers, segmenting audio clips according to classes, or collecting sound files based on similar content. You can also take speech and convert it to text with ease. Audio data requires a few preprocessing steps, including collection and digitization, prior to being ready for analysis by an ML algorithm.

2. Collecting elements and aesthetics



3. Deciding on the color palette and font based on how impactful I want the headings, texts and statements.

Sometimes dark themes improve legibility but lack impact.



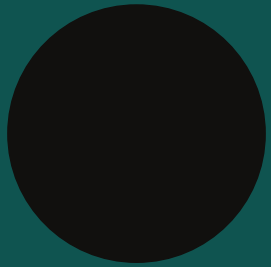
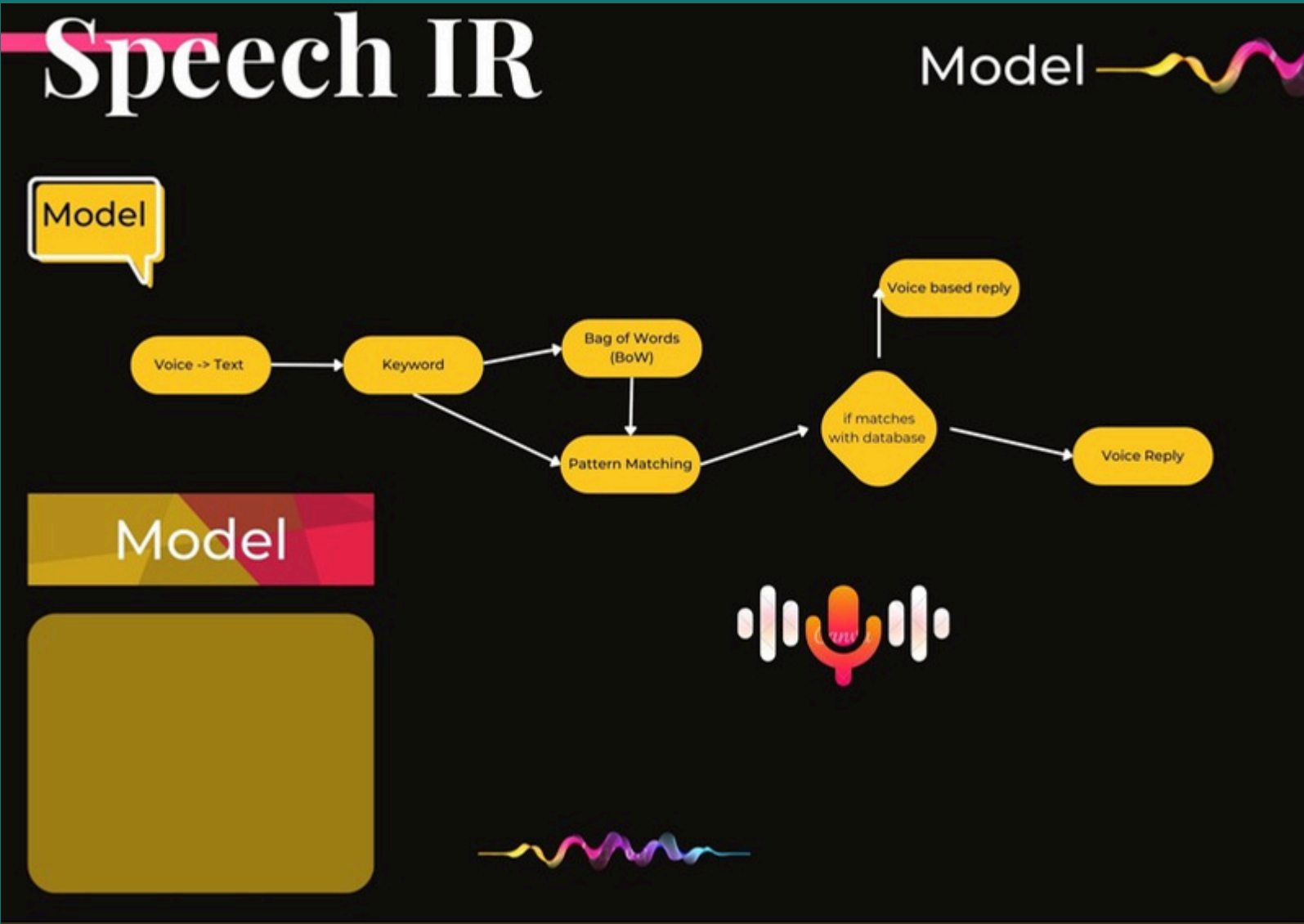
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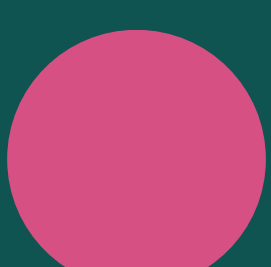
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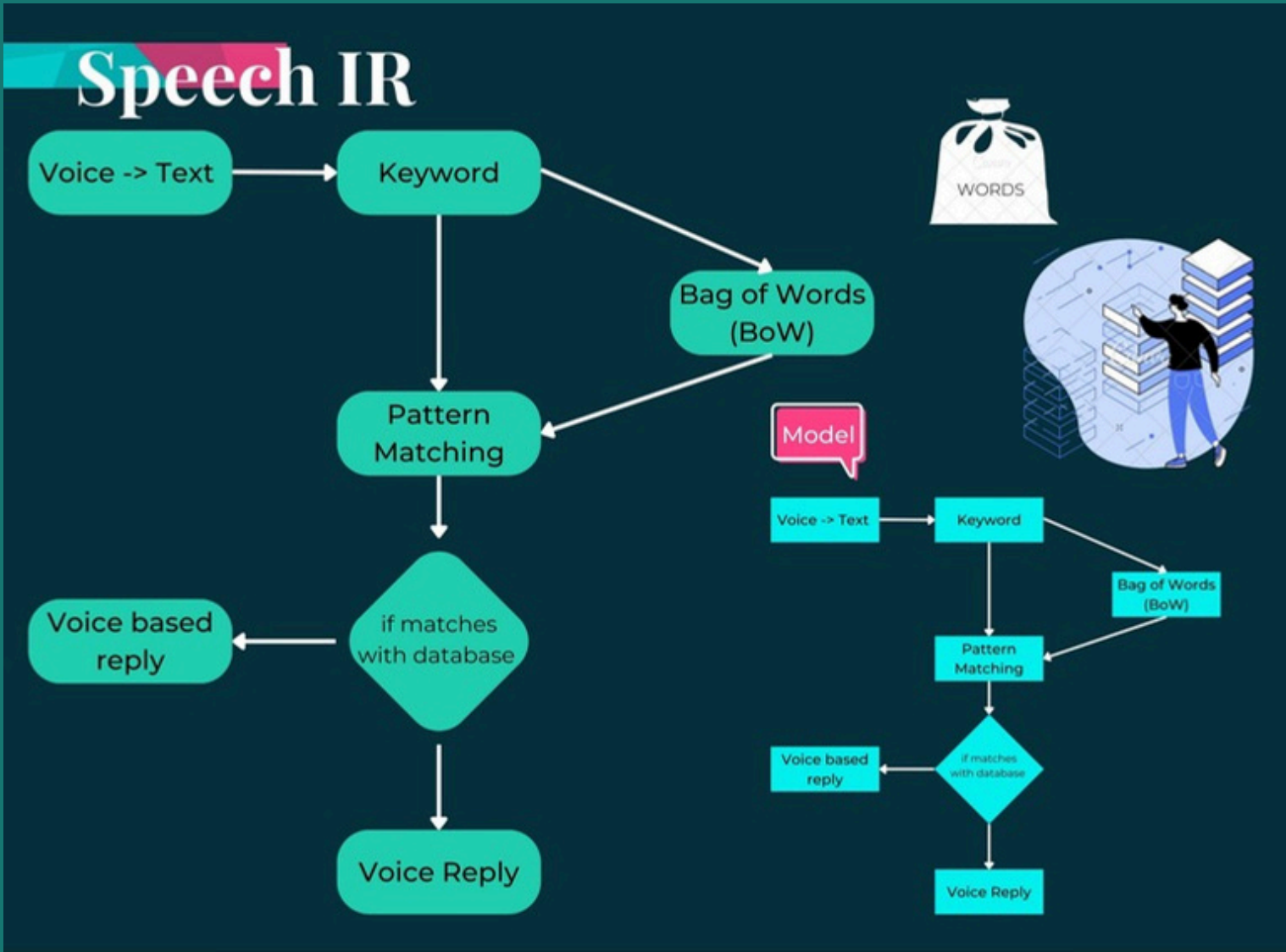
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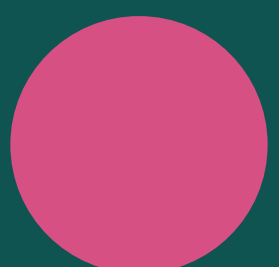
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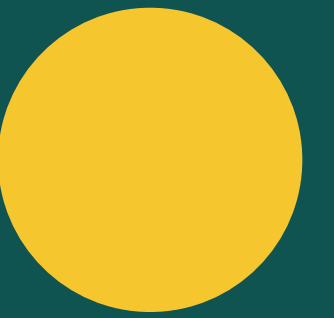
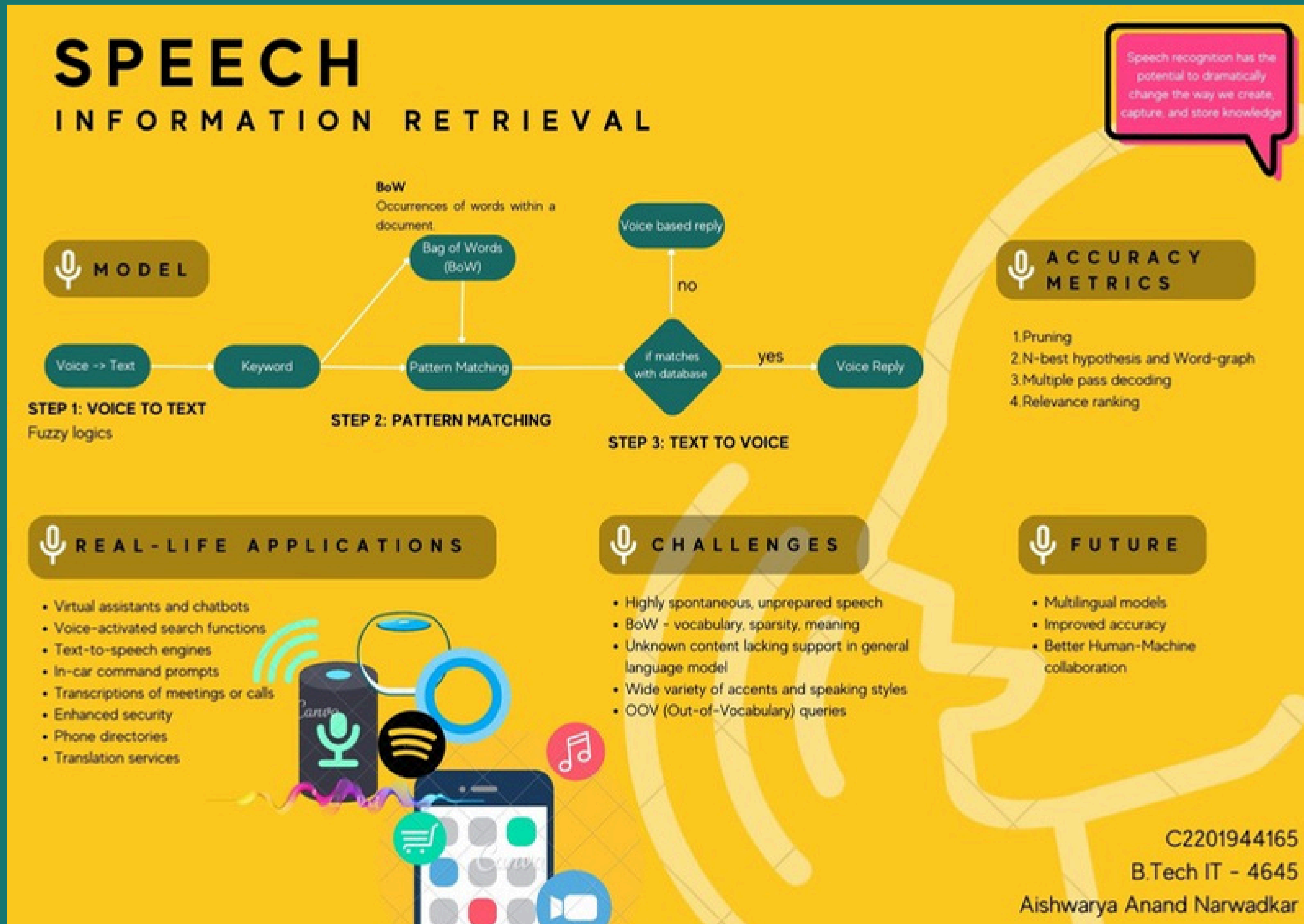


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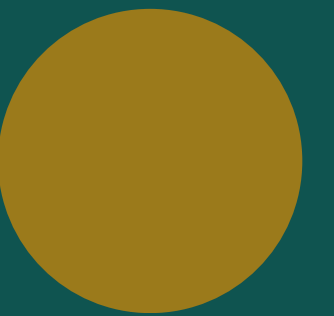
4. Selection and combining poster + context



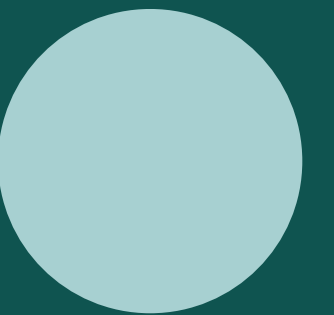
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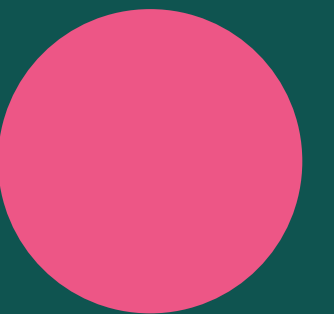
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Speech Information Retrieval

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I. Introduction

Speech Information Retrieval (SIR) is a subfield of Information Retrieval (IR) that deals with the retrieval of information from speech signals. SIR systems are designed to search for specific information contained in audio data, such as transcribed speech, audio archives, and spoken documents.

Key-words: Bag-of-Words, ASR (Automatic Speech Recognition), NLP (Natural Language Processing), DSP (Digital Signal Processing)

II. Methodologies

Accuracy metrics:

- Pruning - pruning tokens for path whose score is low; histograms are used
- N best hypothesis and Word-graph - reports a lattice of n-best scores
- Multiple pass decoding - performs multiple passes for more fine-grained language models
- Relevance ranking - ranking based on relevant scores between segments and queries

Model: Audio data requires a few preprocessing steps, including collection and digitization, prior to being ready for analysis by an ML algorithm.

STEP 1: VOICE TO TEXT

Fuzzy logics applied to spoken sentences based on different accents, pronunciations and speeds. Thus a single word can be represented by a fuzzy set. This model is applied to spoken sentences.

BoW

Occurrences of words within a document. It involves:

- Vocabulary of known words
- Measure of presence of unknown words

STEP 2: PATTERN MATCHING

Boyer-Moore(BM) algorithm can be used which positions the pattern over the leftmost characters in the text and attempts to match it from right to left. If no mismatch occurs then the pattern is found else. The algorithm computes a shift by an amount by which the pattern is moved to the right before a new matching is undertaken. Shift is computed using 2 heuristics: Match and Occurrence heuristic.

STEP 3: TEXT TO VOICE

After getting the text it must analyse and then transform into a phonetic description

NLP Modules:

1. DSP (Digital Signal Processing) modules - It transforms the symbolic information received to audible one as follows: text analysis: first the text is segmented into tokens. The token-to-word conversion creates the orthographic form of the token
2. Application of Pronunciation rules: After the text analysis is completed pronunciation rules can be applied. Silent letters in a word(h in caught) or several phoneme like(m in maximum).

III. Real-Life applications

Today's commercially available ASR models are primarily trained using English-language data sets and consequently exhibit higher accuracy for English-language input.

1. Voice-activated search functions
2. Text-to-speech engines
3. In-car command prompts
4. Transcriptions of meetings or calls
5. Enhanced security
6. Phone directories
7. Translation services

IV. Challenges

ASR faces challenges such as accurate transcription, background noise, language variability, and speaker variability, all of which can interfere with the accuracy of speech recognition and transcription. These challenges make it difficult to develop universal and robust ASR systems capable of transcribing speech in various languages and conditions.

1. Highly spontaneous, unprepared speech
2. Limitations on BoW - vocabulary, sparsity, meaning
3. Unknown content lacking support in general language model
4. Wide variety of accents and speaking styles
5. OOV (Out-of-Vocabulary) queries

V. Future Scope

The next generation of IR tools must provide dramatically better capabilities to learn, anticipate, and assist with the daily information gathering, analysis, and management needs of individuals and groups over time-spans measured in decades.

1. Multilingual models
2. Improved accuracy
3. Better Human-Machine collaboration
4. The next generation of IR tools must provide dramatically better capabilities to learn, anticipate, and assist with the daily information gathering, analysis, and management needs of individuals and groups over time-spans measured in decades.

VI. References

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